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Climate Health and Geological Hazards: Understanding Landslides

Abstract

In January 2026, heavy rain triggered deadly landslides on New Zealand's North Island, including one at a busy campground. ABC News asked climate scientist Kevin Trenberth and emergency management expert Lauren Vinnell to explain how landslides happen, the warning signs to watch for, where they are most common, and how climate change is increasing the risk.

Introduction

Landslides are the downhill movement of rock, soil, and vegetation on a slope. In New Zealand they are called slips. In January 2026, after heavy rain saturated the North Island's east coast, two landslides about 17 kilometres apart struck within days. Two people died when a landslide hit their home, and authorities searched for several others feared dead in a separate slip at a busy campground. One of the landslides occurred at the base of Mauao, Mount Maunganui. The World Health Organization says landslides are the most frequent geological event and can occur anywhere in the world. Understanding what causes them, and how a warming climate changes the risk, can help communities prepare and save lives.

How Do Scientists Study This?

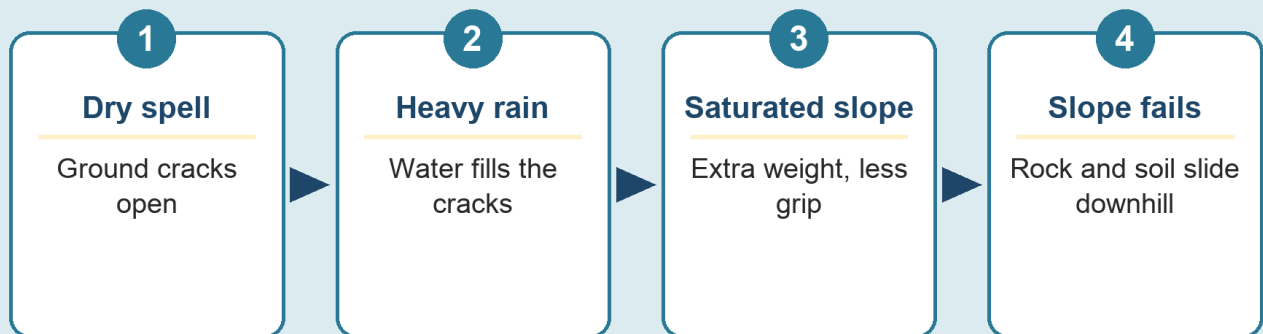


FROM THE ARTICLE

Debris and damage after intense rainfall and floods on New Zealand's North Island. Image via ABC News.

Scientists explain landslides using basic physics: gravity pulls downward on a slope, while the soil, rock, and plant roots provide resistance. When downward forces overcome that resistance, the slope fails. Researchers study how rainfall, dry spells, earthquakes, vegetation removal, and land use change the balance. Emergency managers identify warning signs so people can act before a slide. Climate scientists examine how warming changes rainfall patterns, and health agencies such as the WHO track how many people landslides affect worldwide. Geoscience agencies map where landslides are most likely, based on steep terrain and heavy rainfall.

How climate change sets up landslides



Study the Process

Balance the forces

A slope fails when gravity's downward pull overcomes the resistance of soil, rock, and roots.

Identify triggers

Heavy rain, earthquakes, dry-spell cracks, and vegetation removal can tip the balance.

Map the risk

Steep terrain with heavy rainfall is where landslides are most common.

What Did They Discover?

Trenberth explained that landslides become more likely when soils are saturated after heavy rain. Ironically, a dry spell can set this up by causing cracks: water then infiltrates, adding weight and flowing through the slope until a slab breaks off. Heavy rain also causes erosion, and removing trees destabilizes slopes. Climate change raises the risk through heavier rains and prolonged dry periods, and building plans based on historical records do not account for these extra-heavy rains. Warning signs include falling rocks, small slips, cracks or bulges in the ground, and doors or windows that stick. Between 1998 and 2017, landslides affected about 4.8 million people and caused more than 18,000 deaths worldwide.

Landslides worldwide

4.8M

people affected by landslides, 1998-2017

18,000+

deaths from landslides, 1998-2017

#1

most frequent geological event, according to WHO

Recent disasters show the global scale: at least 1,189 people were killed in floods and landslides across Sumatra, Indonesia, in late 2025.

Takeaways

Gravity versus resistance

Slopes fail when water, cracks, or lost vegetation weaken their hold.

Climate raises risk

Heavier rains and dry spells both make landslides more likely.

Know the signs

Falling rocks, ground cracks, and sticking doors can warn of a slide.

Key Terms

Landslide (slip)

The downhill movement of rock, soil, and vegetation on a slope, driven by gravity.

Saturated soil

Soil whose spaces are completely filled with water, making it heavier and weaker.

Erosion

The wearing away of soil and rock by water, wind, or ice.

Infiltration

Water soaking into the ground, sometimes through cracks, which can destabilize slopes.

Geological hazard

A natural earth process, such as a landslide or earthquake, that can harm people and property.

Activity

Model a Landslide

Students build a slope model to test how water, cracks, and vegetation affect landslide risk.

Materials: Tray or plastic tub; Soil or sand; Watering can or spray bottle; Small plant sprigs or toothpicks as roots; Ruler

1. Build two identical soil slopes in the tray. Add roots to one slope only.
2. Slowly water both slopes and record when and how much soil slides.
3. Rebuild, let one slope dry and press cracks into its surface, then water again. Compare results.
4. Explain your observations using gravity, resistance, saturation, and cracks.
5. Create a warning-signs poster for a community living below steep hills.

References

Bowles, A. (2026). What is a landslide, how do they happen and where are they most common? ABC News, with comments from Kevin Trenberth and Lauren Vinnell.
<https://www.abc.net.au/news/2026-01-23/how-landslides-happen-and-where-they-are-most-common/106261240>